

Module 5: Machine Learning Foundations

Inductive Learning, ID3 Decision Trees (Entropy/Gain), Perceptrons & Backpropagation

Module V: Machine Learning Foundations & Neural Networks

1. Machine Learning Paradigms (Supervised, Unsupervised & Reinforcement Learning)

3. Decision Tree Representation & The ID3 Induction Algorithm

5. Step-by-Step ID3 Calculation Trace on the Benchmark PlayTennis Dataset

7. Overfitting Mitigation: Pre-Pruning vs. Post-Pruning Strategies

9. The Artificial Perceptron Model & The Perceptron Convergence Theorem

11. Multi-Layer Perceptrons (MLP) & Backpropagation Error Gradient Derivations
2. Inductive Learning Hypothesis, Inductive Bias & Occam's Razor Principle

4. Entropy ($H(S)$) & Information Gain Mathematical Formulations

6. C4.5 Algorithm Extensions (Gain Ratio & Continuous Numeric Attributes)

8. Computational Learning Theory: The PAC (Probably Approximately Correct) Model

10. Non-Linear Activation Functions (Sigmoid, Hyperbolic Tangent, ReLU, Softmax)

12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 & 2: Machine Learning Paradigms & Inductive Bias

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E (Tom Mitchell, 1997).

Paradigm	Training Experience (E)	Target Objective (T)	Representative Algorithms
1. Supervised Learning	Labeled training tuples $\langle \mathbf{x}_i, y_i \rangle$ where ground-truth target label y_i is explicitly provided.	Classification (discrete labels) or Regression (continuous real values).	Decision Trees (ID3/C4.5), Support Vector Machines, Neural Networks, Linear Regression.
2. Unsupervised Learning	Unlabeled data points $\mathbf{x}_i$ without ground-truth targets.	Discovering intrinsic clusters, manifold embeddings, and density distributions.	k -Means, Hierarchical Clustering, PCA, Autoencoders, Gaussian Mixture Models.
3. Reinforcement Learning	Sequential interaction with environment receiving scalar reward/penalty feedback r_t .	Learning an optimal action policy $\pi^*(s)$ that maximizes expected discounted cumulative reward.	Q-Learning, SARSA, Deep Q-Networks (DQN), Policy Gradients (PPO).

Inductive Learning Hypothesis & Occam's Razor

- Inductive Learning Hypothesis:** Any hypothesis found to approximate the target function well over a sufficiently large set of training examples will also approximate the target function well over unobserved test examples.
- Occam's Razor:** Prefer the simplest hypothesis that fits the training data (e.g., smaller decision trees are statistically less prone to overfitting than complex deep trees).

Topic 3 – 5: Decision Tree Induction (ID3 Algorithm & Entropy)

1. Information Entropy Formula (Shannon, 1948):

$$H(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

Where p_i is the proportion of examples in set S belonging to target class i . For binary classification (p_+ and p_-):

$$H(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

2. Information Gain Formula:

$$\text{Gain}(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

Where S_v is the subset of S for which attribute A has value v . The ID3 algorithm selects the attribute that maximizes $\text{Gain}(S, A)$ at each node.



Step-by-Step Solved Problem: ID3 Split Calculation on PlayTennis Dataset

Given Dataset: 14 total examples ($|S| = 14$), with 9 Positive ('Yes') and 5 Negative ('No').

Step 1: Compute Overall Dataset Entropy $H(S)$

$$H(S) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = -(0.643 \times -0.637) - (0.357 \times -1.485) = \mathbf{0.940} \text{ bits}$$

Step 2: Evaluate Attribute 'Outlook' (Values: 'Sunny' [2+, 3-], 'Overcast' [4+, 0-], 'Rain' [3+, 2-]):

- $H(S_{\text{Sunny}}) = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = 0.971$
- $H(S_{\text{Overcast}}) = -\frac{4}{4} \log_2(1) - 0 = \mathbf{0.000}$ (Pure leaf!)
- $H(S_{\text{Rain}}) = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) = 0.971$

Step 3: Compute Expected Entropy & Information Gain for 'Outlook':

$$H(S \mid \text{Outlook}) = \frac{5}{14}(0.971) + \frac{4}{14}(0.000) + \frac{5}{14}(0.971) = 0.347 + 0 + 0.347 = 0.694$$

$$\text{Gain}(S, \text{Outlook}) = 0.940 - 0.694 = \mathbf{0.246} \text{ bits}$$

Conclusion: Since 'Outlook' achieves the highest Information Gain among all candidate attributes ('Gain(Wind)=0.048, Gain(Humidity)=0.151, Gain(Temperature)=0.029'), 'Outlook' is selected as the root node!

Topic 9 & 10: Artificial Neural Networks & The Perceptron

The **Perceptron (Rosenblatt, 1958)** is the foundational linear binary classifier model:

$$y = f(\mathbf{w}^T \mathbf{x} + b) = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

Perceptron Learning Rule & Convergence Theorem

$$\mathbf{w} \leftarrow \mathbf{w} + \eta(y_{\text{true}} - y_{\text{pred}})\mathbf{x}$$

Where $\eta \in (0, 1]$ is the learning rate.

Convergence Theorem: If the training dataset is **linearly separable**, the perceptron learning rule is guaranteed to converge to a separating hyperplane in a finite number of weight update steps! If the data is not linearly separable (e.g., the XOR function), single-layer perceptrons fail completely (Minsky & Papert, 1969).

Standard Non-Linear Activation Functions:

Activation Function	Mathematical Equation	Output Range	Derivative $\sigma'(z)$
1. Logistic Sigmoid	$\sigma(z) = \frac{1}{1 + e^{-z}}$	$(0, 1)$	$\sigma(z)(1 - \sigma(z))$
2. Hyperbolic Tangent (tanh)	$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$(-1, 1)$	$1 - \tanh^2(z)$
3. Rectified Linear Unit (ReLU)	$f(z) = \max(0, z)$	$[0, \infty)$	1 if $z > 0$ else 0

Topic 11: Multi-Layer Perceptrons (MLP) & Backpropagation

Backpropagation Gradient Descent Weight Update:

$$w_{ij} \leftarrow w_{ij} - \eta \frac{\partial E}{\partial w_{ij}}$$

Where total sum-of-squares error $E = \frac{1}{2} \sum_k (t_k - y_k)^2$.

Using the Chain Rule of Calculus:

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial \text{net}_j} \frac{\partial \text{net}_j}{\partial w_{ij}} = -\delta_j \cdot x_i$$

- For output neuron k : $\delta_k = (t_k - y_k) \cdot g'(\text{net}_k)$ - For hidden neuron j : $\delta_j = g'(\text{net}_j) \cdot \sum_k w_{jk} \delta_k$

Top BIT Mesra Exam Questions & Answers (Module V)

Q1. Why can a single-layer perceptron not learn the XOR function, and how do Multi-Layer Perceptrons solve this? (8 Marks)

- **Linear Separability Limitation:** The XOR truth table has outputs `(0,0)->0, (0,1)->1, (1,0)->1, (1,1)->0`. Plotting these 4 points on a 2D plane reveals that no single straight line (hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$) can separate class 1 from class 0.

- **Multi-Layer Perceptrons (MLP) Solution:** By adding a hidden layer of non-linear neurons, the network transforms the input space into a higher-dimensional feature space where the points become linearly separable. Two hidden neurons represent intermediate `NAND` and `OR` hyperplanes, which are then combined by an output `AND` neuron.

Q2. Derive the Backpropagation weight update rule for a hidden-to-output weight. (10 Marks)

Let $E = \frac{1}{2}(t_k - y_k)^2$, where $y_k = \sigma(\text{net}_k)$ and $\text{net}_k = \sum_j w_{jk} h_j$.

Applying the Chain Rule:

$$\frac{\partial E}{\partial w_{jk}} = \frac{\partial E}{\partial y_k} \cdot \frac{\partial y_k}{\partial \text{net}_k} \cdot \frac{\partial \text{net}_k}{\partial w_{jk}}$$

1. $\frac{\partial E}{\partial y_k} = -(t_k - y_k)$
2. $\frac{\partial y_k}{\partial \text{net}_k} = y_k(1 - y_k)$ (Sigmoid derivative)
3. $\frac{\partial \text{net}_k}{\partial w_{jk}} = h_j$

Defining error signal $\delta_k = (t_k - y_k)y_k(1 - y_k)$, we get:

$$\frac{\partial E}{\partial w_{jk}} = -\delta_k h_j \implies w_{jk} \leftarrow w_{jk} + \eta \delta_k h_j$$